

Challenge (Habanero)

- Q1.** Rationalize the denominator of $\frac{2}{\sqrt{3}}$
- (A) $\frac{2\sqrt{3}}{3}$
(B) $\frac{2}{3\sqrt{3}}$
(C) $\frac{2\sqrt{3}}{9}$
(D) $\frac{2}{9\sqrt{3}}$
(E) $\frac{4\sqrt{3}}{3}$
- Q2.** A recipe for cooking chicken requires $\frac{1}{\sqrt{2}}$ cups of sauce. Rationalize the denominator to simplify the measurement.
- (A) $\frac{1}{2}\sqrt{2}$
(B) $\frac{\sqrt{2}}{2}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{4}$
(E) $\frac{2}{2\sqrt{2}}$
- Q3.** The height of a building is $\frac{10}{\sqrt{5}}$ meters. Rationalize the denominator to find the height in simplest form.
- (A) $2\sqrt{5}$
(B) $\frac{10}{5}$
(C) $\frac{10\sqrt{5}}{5}$
(D) $\frac{2}{\sqrt{5}}$
(E) $5\sqrt{2}$
- Q4.** The area of a triangle is given by $\frac{1}{2}bh$, where b is the base and h is the height. If the base is $2\sqrt{3}$ meters and the height is $\frac{4}{\sqrt{3}}$ meters, find the area of the triangle with a rationalized denominator.
- (A) $\frac{4}{3}$
(B) $\frac{4\sqrt{3}}{3}$
(C) $\frac{8}{3}$
(D) $2\sqrt{3}$
- (E) $\frac{8\sqrt{3}}{3}$
- Q5.** The length of a rectangular room is $\frac{6}{\sqrt{2}}$ meters. Rationalize the denominator to simplify the measurement.
- (A) $3\sqrt{2}$
(B) $\frac{6}{2}$
(C) $\frac{6\sqrt{2}}{2}$
(D) $2\sqrt{3}$
(E) $\frac{3}{\sqrt{2}}$
- Q6.** The recipe for a cake requires $\frac{3}{\sqrt{5}}$ cups of flour. Rationalize the denominator to simplify the measurement.
- (A) $\frac{3\sqrt{5}}{5}$
(B) $\frac{3}{5}$
(C) $\frac{\sqrt{5}}{3}$
(D) $\frac{5}{3}$
(E) $\frac{15}{\sqrt{5}}$
- Q7.** The width of a picture frame is $\frac{8}{\sqrt{6}}$ inches. Rationalize the denominator to simplify the measurement.
- (A) $\frac{4}{\sqrt{3}}$
(B) $\frac{8\sqrt{6}}{6}$
(C) $\frac{8}{6}$
(D) $\frac{4\sqrt{3}}{3}$
(E) $\frac{8}{3}$
- Q8.** The height of a skyscraper is $\frac{50}{\sqrt{10}}$ meters. Rationalize the denominator to find the height in simplest form.
- (A) $5\sqrt{10}$
(B) $\frac{50}{10}$
(C) $\frac{50\sqrt{10}}{10}$
(D) $\frac{5}{\sqrt{10}}$
(E) $25\sqrt{2}$

Open Ended Questions (FRQ)

- Q9.** A construction worker needs to mix $\frac{2}{\sqrt{7}}$ bags of cement with $\frac{3}{\sqrt{5}}$ bags of sand. Rationalize the denominators and simplify the total amount of mixture required.
- Q10.** The area of a plot of land is given by $\frac{1}{2} \cdot \frac{10}{\sqrt{3}} \cdot \frac{8}{\sqrt{2}}$. Rationalize the denominators and simplify the expression to find the area of the land.

Chef's Tips

- Remind students to simplify their answers.
- Encourage students to use calculators to check their work.
- Emphasize the importance of rationalizing denominators in real-world applications.